Resonant cascaded Kerr nonlinearity in $\chi^{(2)}$ metasurfaces

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Abstract: Cascaded second-harmonic and difference-frequency generation produces an effective Kerr response whose strength in bulk crystals is set by the phase mismatch. We show that a second-harmonic resonance detuned by $\Delta\omega$ from twice the pump frequency takes the place of the phase mismatch, and derive the resulting susceptibility in closed form. It changes sign with $\Delta\omega$, carries a nonlinear loss, and obeys a bandwidth sum rule. In $LiNbO_3$ it exceeds the native Kerr susceptibility by about 50 in a doubly resonant metasurface and about 10^3 in a poled microring. We propose a picosecond Z-scan to detect the sign change in a singly resonant metasurface.

1. Introduction

1. Kerr nonlinearity underlies all-optical switching, optical bistability, parametric oscillation and frequency combs. The native $\chi^{(3)}$ of low-loss dielectrics is small, about 2×10^{-21} m^2/V^2 in $LiNbO_3$, and field enhancement in resonant nanostructures increases the nonlinear phase per unit power without changing the material susceptibility.

2. Two cascaded second-order processes act on the pump as a Kerr medium. The effect was predicted by Ostrovskii [1], observed in KTP as a nonlinear refraction that changes sign with the phase mismatch [2], reviewed in Ref. [3], and described microscopically as a collective effect [4]. In bulk crystals its strength scales as $1/\Delta k$, and it is used for soliton compression [5, 6].

3. Metasurfaces with bound states in the continuum enhance second-harmonic generation [7–11], and cascaded second-order processes in metasurfaces have produced third-harmonic light [12], while the back-conversion of the second harmonic has been treated as pump depletion [13]. In $\chi^{(2)}$ microresonators the cascaded Kerr response has been tuned through the detuning of the second-harmonic mode [14, 15]. We are not aware of a treatment of the metasurface cascade as a Kerr response on the pump that compares it with the native response of the host.

4. Here a detuned second-harmonic resonance replaces the phase mismatch, so that the cascaded susceptibility scales as $Q_2/\Delta\omega$ and changes sign with $\Delta\omega$. A single expression covers bulk crystals, metasurfaces and microrings, and with it we identify the hosts and resonators in which the cascade exceeds the native Kerr response. For $LiNbO_3$ the factor is about 50 in a doubly resonant metasurface and about 10^3 in a poled microring, within a bandwidth set by the second-harmonic linewidth.

2. Cascaded susceptibility at a second-harmonic resonance

With real fields written as $\mathbf{E}e^{-i\omega t}$ + c.c., the second-harmonic and difference-frequency polarisations are $\varepsilon_0\chi^{(2)}\mathbf{E}_1\mathbf{E}_1$ and $2\varepsilon_0\chi^{(2)}\mathbf{E}_2\mathbf{E}_1^*$, and self-phase modulation is $3\varepsilon_0\chi^{(3)}|\mathbf{E}_1|^2\mathbf{E}_1$ [16]. The

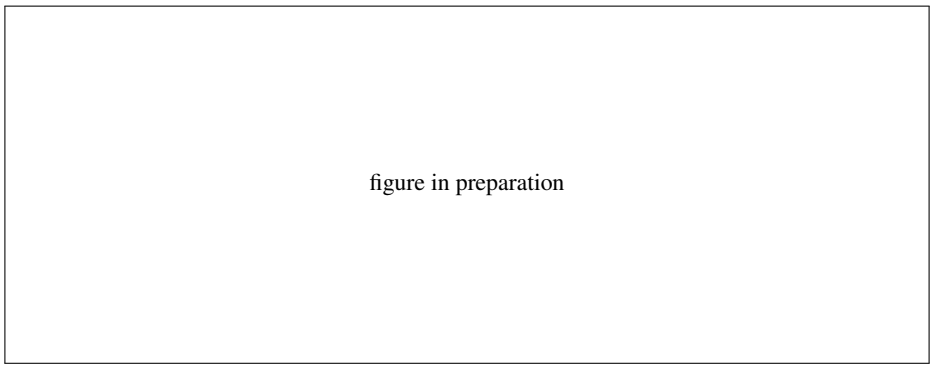


Fig. 1. Cascaded Kerr response at a second-harmonic resonance. (a) Second-harmonic generation and difference-frequency generation through a mode at ω_2, detuned by $\Delta\omega = 2\omega_1 - \omega_2$. (b) Doubly resonant $LiNbO_3$ metasurface with a quasi-bound state at ω_1 and a symmetry-protected mode near $2\omega_1$ in one meta-atom. (c) Periodically poled microring with pump and second-harmonic modes. (d) Real (solid) and imaginary (dashed) parts of $\chi^{(3)}_{\mathrm{casc}}$ versus $\Delta\omega/\gamma_2$, normalised to the value at $\Delta\omega = 2\gamma_2$.

second-harmonic field driven by $\varepsilon_0\chi^{(2)}\mathbf{E}_1\mathbf{E}_1$ is written with the Green's function of the structure, whose pole term dominates near an isolated quasinormal mode [17, 18]. The same result follows from coupled-mode equations for the pump and second-harmonic modes [19], in which the second-harmonic amplitude follows the pump algebraically in steady state. Both derivations give

$$\chi^{(3)}_{\mathrm{casc}} = -\frac{2\varepsilon_0\omega_1}{3N_2}\,\frac{|M|^2}{(\Delta\omega + i\gamma_2/2)\int |\mathbf{f}_1|^4\, d^3r} \simeq -\frac{2\omega_1}{3n_2^2\Delta\omega}\,|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}, \tag{1}$$

with $\mathbf{f}_{1,2}$ the mode profiles, $M = \int \chi^{(2)}_{ijk} f^*_{2,i} f_{1,j} f_{1,k}\, d^3r$, N_2 the norm of the second-harmonic mode, γ_2 its energy decay rate and n_2 its effective index, and the geometric overlap

$$\eta_{\mathrm{geom}} = \frac{\varepsilon_0 n_2^2 |M|^2}{|\chi^{(2)}|^2 N_2 \int |\mathbf{f}_1|^4\, d^3r} \le 1.$$

The derivations are given in Supplement 1.

For phase-matched plane waves in a uniform medium, $\eta_{\mathrm{geom}} = 1$ and Eq. (1) becomes the bulk bound-wave susceptibility $-2|\chi^{(2)}|^2/[3(n_2^2 - n_1^2)]$, with the frequency mismatch $\Delta\omega$ in the role of the wavevector mismatch Δk. The expression assumes one second-harmonic mode, $|\Delta\omega| \gtrsim \gamma_2$ and an undepleted pump, and a pump bandwidth small compared with $|\Delta\omega + i\gamma_2/2|$. Cascaded optical rectification with the electro-optic effect [20] and the longitudinal part of the bound second harmonic add a background that is not resonantly enhanced.

3. Limits and host materials

The imaginary part of $\chi^{(3)}_{\mathrm{casc}}$ is even in $\Delta\omega$ and the real part is odd; their ratio $\rho = \gamma_2/(2|\Delta\omega|)$ is $1/4$ at $\Delta\omega = 2\gamma_2$, and bistability and parametric gain both require $\rho < 1/\sqrt{3}$. The peak susceptibility grows as Q_2 and its bandwidth falls as γ_2, so that $|\chi^{(3)}_{\mathrm{casc}}|_{\max}\,\Delta f \lesssim \omega_1|\chi^{(2)}|^2\eta_{\mathrm{geom}}/(3\pi n_2^2)$ independently of Q_2 [Fig. 2(a)]. Trade-offs between strength and bandwidth of resonant nonlinear enhancement have been discussed for plasmonic and native nonlinearities [21], for harmonic generation [22], and for a cavity-enhanced cascaded frequency shifter [23]. Pump depletion sets in when the second-harmonic coupling rate approaches $|\Delta\omega|$, beyond which the steady state becomes parametrically unstable [24].

Phase-mismatched bulk PPLN gives about 10^{-20} m^2/V^2 over 10 THz, and a doubly resonant metasurface with $Q_2 = 4000$ gives about 10^{-19} m^2/V^2 over 100 GHz, so bulk cascading is ahead in $|\chi^{(3)}|\,\Delta f$ by about an order of magnitude while the metasurface uses a sub-micron layer and continuous-wave pumping.

The enhancement over the native response, $R = |\chi^{(3)}_{\mathrm{casc}}|/|\chi^{(3)}_{\mathrm{nat}}| = Q_2|\chi^{(2)}|^2\eta_{\mathrm{geom}}/(6n_2^2\chi^{(3)}_{\mathrm{nat}})$ at $\Delta\omega = 2\gamma_2$, depends on the host through $|\chi^{(2)}|^2/(n_2^2\chi^{(3)}_{\mathrm{nat}})$. With $\eta_{\mathrm{geom}} = 0.3$, $R = 1$ is reached at $Q_2 \approx 80$ in $LiNbO_3$, near 2×10^3 in GaP, AlN and GaAs, and near 2×10^4 in AlGaAs, whose native response is enhanced near the band edge [Fig. 2(b); parameters in Supplement 1].

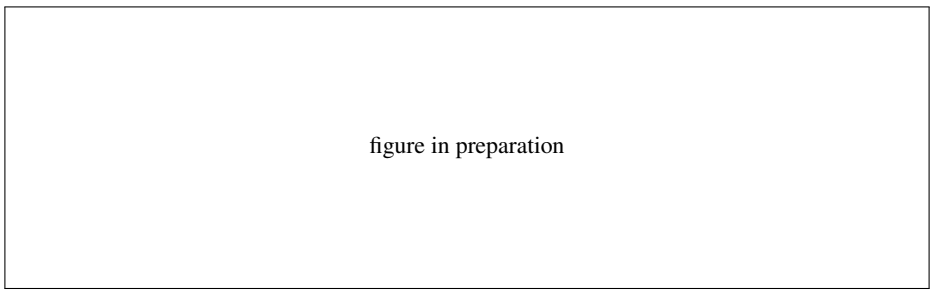


Fig. 2. (a) Peak $|\chi^{(3)}_{\mathrm{casc}}|$ (circles, left axis) and its bandwidth (squares, right axis) versus Q_2 for $LiNbO_3$ at $\Delta\omega = 2\gamma_2$ and $\eta_{\mathrm{geom}} = 0.3$. (b) R versus Q_2 at $\Delta\omega = 2\gamma_2$ and $\eta_{\mathrm{geom}} = 0.3$ for $LiNbO_3$, GaP, AlN, GaAs and AlGaAs (line styles); triangles mark $R = 1$.

4. Metasurface implementation

4.1. Doubly resonant metasurface

A symmetry-protected bound state in the continuum near $2\omega_1$ has a quality factor limited by absorption and fabrication [8], and a quasi-bound state at ω_1 couples the pump from free space [7, 9]; doubly resonant cavities of this kind have been used for second-harmonic generation [25, 26]. The overlap M is nonzero only if the unit-cell symmetry of $\mathbf{f}_2$ matches that of $\chi^{(2)}$ contracted with $\mathbf{f}_1\mathbf{f}_1$, which sets selection rules for the d_{33} orientation of a $LiNbO_3$ film [11]. At normal incidence both modes lie at the Γ point, which removes the bulk phase-matching condition, and $\Delta\omega$ is tuned by temperature or by the electro-optic effect. Disorder and finite array size lower the collective quality factor of the second-harmonic mode, and we estimate both from supercell simulations.

With a common pump resonance the cascaded and native architectures share the field enhancement, and each observable scales with a fixed power of $1 + R$: self-phase modulation as $1 + R$, the bistability threshold and switching energy as $1/(1 + R)$, and four-wave mixing as $(1 + R)^2$ [Fig. 3]. For a $LiNbO_3$ design with $Q_2 = 4000$, $\eta_{\mathrm{geom}} = 0.3$ and $\Delta\omega = 2\gamma_2$, $R \approx 49$. Cascaded bistability has been observed in a monolithic $LiNbO_3$ resonator with a phase-mismatched second harmonic [27].

4.2. Singly resonant metasurface and proposed measurement

Without a pump resonance the pump field in the meta-atom is fixed by the incident wave. The overlap becomes $\widetilde{M} = \int \chi^{(2)}_{ijk} f^*_{2,i} e_{1,j} e_{1,k}\, d^3r$, with $\mathbf{e}_1$ the linear pump field per unit incident amplitude, and the unit-cell volume replaces $\int |\mathbf{f}_1|^4$ in Eq. (1). The per-photon phase loses the Q_1^2 factor of the pump resonance and bistability is absent, so this structure serves as a measurement of the cascaded susceptibility. We evaluate one fabricable $LiNbO_3$ design with

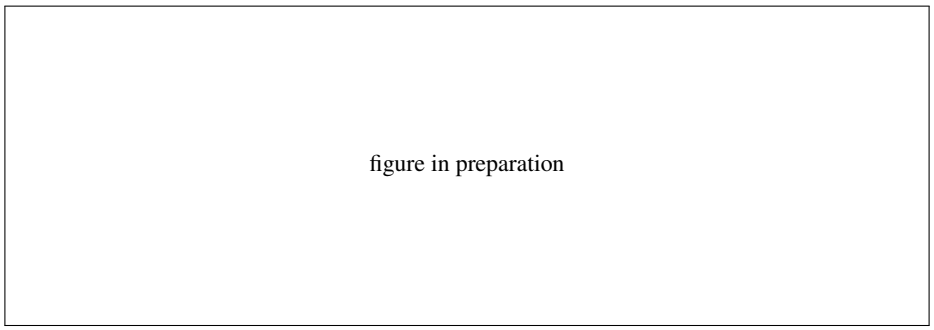


Fig. 3. Doubly resonant $LiNbO_3$ metasurface. (a) Unit cell with the pump and second-harmonic mode profiles. (b) R versus Q_2 and η_{geom}, with the design point marked. (c) Stored pump energy versus input power for increasing and decreasing sweeps, with (solid) and without (dashed) the second-harmonic mode, from two-mode coupled-mode simulations at $\Delta\omega = -2\gamma_2$ and $\gamma_1 = 10\gamma_2$. (d) Ratio of four-wave-mixing efficiencies with and without the second-harmonic mode, with $(1 + R)^2$ as a dashed line.

full-wave simulations, compute $\widetilde{M}$ from its fields, and give $\chi^{(3)}_{\mathrm{casc}}(\Delta\omega)$, R and the single-pass phase; designs simulated so far reach R between 2 and 10.

The sign change would be measured with a closed-aperture Z-scan as a function of pump wavelength, which moves $2\omega_1$ across ω_2, in analogy with the angle-tuned measurement in KTP [2]. The pulse must be longer than the second-harmonic lifetime $1/\gamma_2$, and its bandwidth must stay below about $2\gamma_2$. Near a Mie resonance the native response alone can change the sign of the effective nonlinear index [28]; the cascaded part is the component odd in $\Delta\omega$ at fixed pump detuning, which is well defined when the pump transmission is flat across the scan. The second-harmonic conversion efficiency of the same sample would fix the absolute value of $\widetilde{M}$.

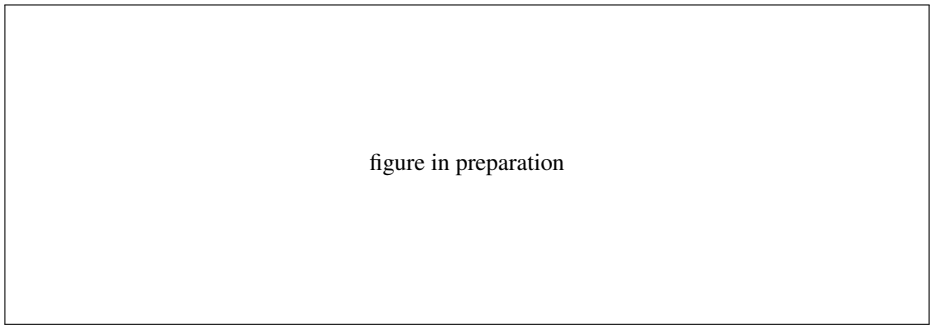


Fig. 4. Singly resonant $LiNbO_3$ metasurface. (a) Unit cell and second-harmonic mode profile. (b) Real (solid) and imaginary (dashed) parts of $\chi^{(3)}_{\mathrm{casc}}$ versus $\Delta\omega/\gamma_2$. (c) Simulated closed-aperture Z-scan transmission for $\Delta\omega = \pm 2\gamma_2$ (solid and dashed). (d) Z-scan phase for picosecond pulses, relative to its continuous-wave value, versus pulse duration.

5. Microring resonators

The same elimination applies to a doubly resonant $\chi^{(2)}$ microring, where it reproduces the cavity expression of Refs. [14, 15], and the measured second-harmonic efficiency fixes the coupling rate [29–31]. For periodically poled thin-film $LiNbO_3$, $R \approx 2.8 \times 10^3$ for the device of Ref. [30] and about 13 without poling. Since $Q_2 < 2Q_1$, the second-harmonic linewidth exceeds the pump

linewidth and the modulation bandwidth is reduced by a few percent. Bistability and parametric oscillation move from about 14 mW and 9 mW for the native response of the same ring to about 35 μW and 8 μW [Fig. 5]. Thermal and photorefractive shifts [32] are comparable to the cascaded shift at these powers; to leading order they are even in $\Delta\omega$, and an electro-optic dither of $\Delta\omega$ with lock-in detection at zero pump detuning would separate them from the odd cascaded shift. The ring supports continuous-wave switching at femtojoule intracavity energy, cancellation of its native Kerr response, and a choice of the sign of the Kerr term for modulation instability and combs [33, 34].

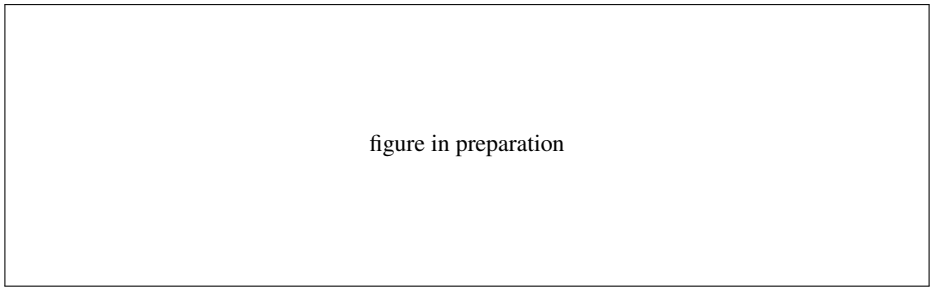


Fig. 5. Periodically poled $LiNbO_3$ microring. (a) R versus Q_2 at $\Delta\omega = 2\gamma_2$ for poled (solid) and unpoled (dashed) rings; the star marks the device of Ref. [30]. (b) Transmission at 30 μW input for $\Delta\omega = +2\gamma_2$ (solid), $\Delta\omega = -2\gamma_2$ (dashed) and the native response (dotted). (c) Bistability (circles) and parametric (squares) thresholds versus Q_2, cascaded (filled) and native (open).

6. Discussion and conclusion

Table 1. Cascaded Kerr Response on Four Platforms (R at $\Delta\omega = 2\gamma_2$)

Platform	Set by	Q_2	R	Bandwidth	CW device
Bulk PPLN	Δk	–	≈ 5	≈ 10 THz	no
Doubly resonant metasurface	$\Delta\omega$	4×10^3	≈ 50	≈ 100 GHz	marginal
Singly resonant metasurface	$\Delta\omega$	10^3–6×10^3	2–10	0.1–0.4 THz	no
Poled $LiNbO_3$ microring	$\Delta\omega$	3×10^5	$\approx 2 \times 10^3$	≈ 0.2 GHz	yes

The platforms in Table 1 differ in what sets the cascade and in the bandwidth that remains. Engineered second-order media raise R through $|\chi^{(2)}|^2$, as in coupled quantum wells with an interband $\chi^{(2)}$ above 10^3 pm/V [35] and quantum-well metasurfaces [36], provided the native third-order response, which is resonantly enhanced in intersubband systems [37], stays small. A detuned second-harmonic resonance turns the cascaded second-order response into a Kerr nonlinearity of either sign whose strength in $LiNbO_3$ metasurfaces and microrings exceeds the native response by one to three orders of magnitude.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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